

system. This helps determine and fix any SI issue even before the PCB is manufactured. Physical design mainly involves the PCB layout based on the schematics and net-list provided by the logical design team and SI related inputs provided by the electrical design team (such as the trace lengths and component placement for better SI). A good SI design also ensures less electromagnetic radiation and therefore a good and inexpensive design for lower EMI.

For beginners, these are some guidelines to achieve an acceptable overall system design. Following these tips would help avoid design deadline slips and minimise debugging of board-level, software-level or system-level issues.

Hardware board design

Consult the evaluation platform. Evaluation platforms are not only meant for application development and testing but these also act as good reference design sources. Logical designers are likely to discover and hence later reuse a lot of interfaces and components for the design. The most useful ones are the memory and input/output (I/O) interfaces such as audio-video codecs, data converters and power monitoring components. A lot that is present here could be reused if referred during the early phases of design.

The biggest advantage is that these circuits are already tested by evaluation platform manufacturers and therefore are not required to be validated to prove the concept of interfacing, etc. For example, one could use the same power sequencing/monitoring scheme as being implemented on the evaluation platform.

Anomaly/errata documents. Nothing is perfect and so are the ICs and other components used in circuit design. Before working on any logical design, it is important to read and understand the anomaly or errata sheets for a specific version of the IC carefully. Most defects have either software or hardware workarounds but there could be some rare cases which do not have a work-

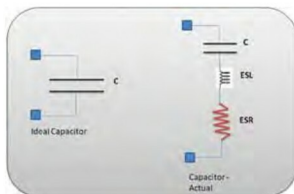


Fig. 1: Ideal model (left) and practical model (right) of a capacitor

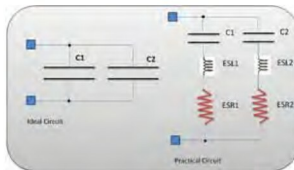


Fig. 2: Parallel circuit with ideal and practical models of capacitors

around. Then it is better to avoid those situations by not using those specific features or conditions. Some anomalies may have only hardware workarounds and therefore, considering hardware board design cost and time, designing a PCA board without the knowledge of such issues could turn out to be the worst-ever nightmare.

Silicon (IC) revisions, future roadmap and last-time buy. Always work with the latest silicon version available at the time. This would most likely have most of the fixes for known anomalies. Find out from the technical support team whether the chip manufacturer is planning to release any latest silicon revision and how that would differ from the existing one in production.

For older ICs, consult the marketing team and know about the last-time-buy date for those ICs. You wouldn't want to get into a situation where your company starts selling the products and the manufacturers of certain ICs stop their production. This is very like-

ly to happen with old/legacy parts. A solid future roadmap ensures smoother migration to newer parts of the same family, hence ensuring code/instruction set compatibility and at times a drop-in replacement part with enhanced feature set.

Bypass/decoupling capacitors for ICs. Most complex ICs, such as DSPs and microprocessors and ICs in which a lot of signal switching happens, need bypass/decoupling capacitors between V_{DD} (supply) pin and GND (ground) pin, as close as possible to these pins. Though there will be additional board design cost and real estate concerns, ensure that the design

makes use of sufficient number of decoupling capacitors and an appropriate decoupling scheme.

Practically, in a system, the power supply would be physically distanced from these ICs and there will be tracks running from the output of the power supply to power pins of ICs. Under static (steady state) condition, there is no problem and all supply voltage levels at the pins of the IC look good. The problem arises when a lot of switching happens inside the IC. This happens when the core starts execution and I/Os start toggling.

Under these dynamic conditions, power supply tracks behave as inductors whose fundamental property is to resist any change in current ($\Delta i/\Delta t$) through it. The switching of internal transistors demands more dynamic current but this $\Delta i/\Delta t$ generates back-e.m.f. (electromotive force = $L \times (\Delta i/\Delta t)$ volts) across inductive traces. Due to the lack of dynamic power required by the processor, the application running on the processor could crash. Ad-